Instance Segmentation of Solar Filaments with Mask R-CNN: Reaching Human-Level Agreement on the MAGFiLO Benchmark

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Abstract

We address the Solar Filament Segmentation Challenge 2026, which asks for one pixel-precise mask per filament in 2048 × 2048 H-alpha images, scored by Panoptic Quality (PQ). We fine-tune a COCO-pretrained Mask R-CNN (ResNet50-FPN) with flip augmentation at 1280 × 1280 input, tune the score threshold and minimum mask size directly on PQ, and make predicted masks disjoint as the evaluation requires. The final model reaches a PQ of 0.36 on the public leaderboard (0.393 on our validation split) and runs on a 6 GB laptop GPU. Scoring one annotator's masks against another's on the same image gives a PQ of only 0.34, so our model's outlines are about as consistent with the labels as human outlines are with each other. An error analysis shows that recall for small filaments, not precision, is the main remaining weakness.

1 Introduction

Solar filaments are dense, cool structures suspended in the corona that appear as dark ribbons in H-alpha images. Their eruptions are linked to coronal mass ejections and space weather, yet no operational system currently provides continuous automated filament detection. The challenge asks for pixel-precise instance segmentation of filaments in ground-based images from the MAGFiLO dataset [1]. It scores Panoptic Quality [2] and penalizes fragmenting one filament into several predictions, merging several filaments into one, and computational cost.

We use a pretrained two-stage instance segmentation model, Mask R-CNN [3], tuned on PQ rather than on training loss. Our contributions are modest and practical: (i) a complete, reproducible pipeline that produces valid submissions, including a disjoint-mask post-processing step; (ii) an error analysis by failure type and filament size; and (iii) a measurement of inter-annotator agreement that puts the achievable score in context.

2 Data

The training data are 2048 × 2048 grayscale H-alpha images from the GONG network. Each annotation is one polygon for one filament, with a category label encoding chirality, which we do not use. Following the organizers' instruction not to deduplicate by file name, repeated annotator passes of the same image are kept as separate samples, giving 1,154 annotated samples. We hold out 10% of the *files* (116 samples) for validation, split by file name with a fixed seed so that no image appears in both training and validation. The test set contains 180 images.

Polygons are rasterized to binary masks with `pycocotools`, boxes are derived from the masks, and empty masks are discarded. Images are scaled to $[0, 1]$, replicated to three channels to match the pretrained backbone, normalized with ImageNet statistics, and resized inside the detector to the network input size (1280 × 1280 for the final model, 1024 × 1024 for the baselines). Predictions are mapped back to the full 2048 × 2048 grid. During training we apply random horizontal and vertical flips.

3 Method

Model. We use torchvision's Mask R-CNN with a ResNet50-FPN backbone [4, 5] initialized from COCO weights [6]. The box and mask predictors are replaced by two-class versions (background, filament); chirality is not predicted, since the metric scores instance masks only.

Training. We train with SGD (momentum 0.9, learning rate 5×10^{-3}, weight decay 10^{-4}), a linear warmup followed by cosine decay, batch size 1, mixed precision, and gradient-norm clipping at 10, for 16 epochs at 1280 px (20 epochs at 1024 px for the baseline). One epoch takes about 5.6 minutes at 1280 px (4.5 at 1024 px) on a single RTX 4050 laptop GPU with 6 GB of memory.

Post-processing. At inference we keep predictions with score ≥ 0.8, binarize masks at 0.5, and drop masks smaller than 200 pixels (0.9 and 50 for the 1024 px baseline). The evaluation rejects images whose predicted masks overlap, and Mask R-CNN predicts instances independently, so we make the masks disjoint: instances are processed from highest to lowest score, and each keeps only pixels not already claimed. Because priority follows score, removing low-score instances never changes the pixels of higher-score ones, which keeps threshold sweeps on cached predictions exact. The threshold and minimum area were chosen by grid search maximizing local PQ on the validation split.

Model	Local PQ	Public PQ
12 epochs, no augmentation	0.357	0.34
+ flips, 20 epochs (epoch 17)	0.389	0.34
+ 1280 px input, 16 epochs	0.393	0.36

Table 1: Local validation and public leaderboard PQ. Re-running epochs 17-20 gave 0.380-0.387 locally, i.e. no meaningful change. Epochs 12, 14 and 16 of the 1280 px run scored 0.388, 0.396 and 0.393; the row reports the final epoch.

Evaluation. Our local metric reproduces the organizers' notebook: a prediction matches a ground-truth filament when their IoU exceeds 0.5, and true positives, false positives and false negatives are pooled over all images,

$$\mathrm{PQ} = \frac{\sum_{(p,g)\in \mathrm{TP}} \mathrm{IoU}(p,g)}{|\mathrm{TP}| + \frac{1}{2}|\mathrm{FP}| + \frac{1}{2}|\mathrm{FN}|}. \quad (1)$$

Masks are encoded as compressed run-length encodings with `pycocotools`.

4 Results

Models and scores. Table 1 lists the models we evaluated. Local PQ is computed on our validation split after making masks disjoint; the public score is the leaderboard value. For each model the score threshold and minimum area were picked by grid search on the validation split.

Higher input resolution. Training the same recipe at 1280×1280 for 16 epochs gave a local PQ of 0.393 (0.388-0.396 across epochs 12-16, score threshold 0.8) against 0.389 at 1024 px. The model finds more filaments (523 versus 475 true positives) but also makes more false alarms (312 versus 198), so PQ barely changes; the difference is within the noise of our validation split.

Further experiments. We tested three ideas aimed at the weaknesses in the error analysis (Table 2). Training only on filaments that at least two annotators drew (consensus labels) lowered false alarms (288 versus 312) but also matches (503 versus 523). Replacing the default anchors with smaller and more elongated ones (16-256 px, ratios 1:4 to 4:1) raised the share of small filaments with a well-fitting anchor from 54% to 99%, yet the number of matches stayed at 523 and false alarms rose to 339. Finally, none of the 834 predictions on the validation split lies outside the solar disk, so restricting predictions to the disk would change nothing. None of these changes improved on the plain 1280 px model, and the differences are within the noise of our validation split.

Inference cost. Predicting all 180 test images, including image loading and RLE encoding, takes 35 s (0.19 s per image) with the 1024 px model and 52 s (0.29 s per image) with the final 1280 px model, on the laptop GPU.

1280 px variant	PQ	TP	FP	FN
Final model (flips)	0.393	523	312	385
+ consensus labels	0.388	503	288	405
+ small anchors	0.387	523	339	385

Table 2: Local validation results at epoch 16, each at its best score threshold.

Ground-truth filaments (908)	Share
Found (best IoU > 0.5)	52%
Near miss (IoU 0.3-0.5)	10%
Weak overlap (IoU 0.1-0.3)	4%
No overlapping prediction	34%

Table 3: Fate of each validation filament for the epoch-17 model (score ≥ 0.9).

Error analysis. Table 3 breaks down the epoch-17 model on the 908 validation filaments. It finds 52% of them, and for 34% it produces no overlapping prediction at all. Only 72 of its 672 predictions overlap no filament, and the mean IoU of hits is 0.647. Recall is therefore the main weakness, especially for small filaments: the hit rate rises from 35% for the smallest quarter (< 644 pixels) to 53%, 64% and 57% for the next three quarters.

Human agreement. The training set contains several annotation passes for the same image. On the 296 files with at least two passes, scoring one annotator's masks against another's with the same PQ gives 0.340 (mean IoU of matched filaments 0.634), and annotators of the same image differ by 2.6 filaments on average. Our model's matched filaments have a mean IoU of 0.647.

5 Discussion

Our final model reaches a local PQ of 0.393 and a public PQ of 0.36, similar to or slightly above the 0.34 agreement we measure between two annotators of the same image. Part of the misses are therefore likely filaments that one annotator drew and another did not, which no model can match against every annotator. This is an interpretation: the measurement is pairwise and made on training files, and a model trained on several annotators can legitimately score above pairwise human agreement.

Local and public scores did not always move together. Augmentation and longer training raised the local score from 0.357 to 0.389 while the public score stayed at 0.34. Training at 1280 px left the local score within noise (0.393) but raised the public score to 0.36. We cannot separate several explanations: the validation split is small and noisy; we selected checkpoints and thresholds on it, which biases the local score up-

ward; and the public score may be computed on a subset of the test set. The 1280 px model finds more filaments (523 versus 475 matches) at the cost of more false alarms (312 versus 198), and the error analysis shows that small filaments are the weakest group, so better recall on them is a plausible reason for the public gain. With one public score per configuration, we do not claim it as a robust improvement. Two attempts to act on suspected causes of the remaining errors, label consensus and anchor shapes, did not raise PQ; we did not test directly whether annotation ambiguity is the limiting factor. Following the organizers' guidance, we do not compare our score with other teams'.

6 Limitations

- The validation split holds 10% of the files (116 samples), and the score threshold, minimum area and checkpoint were all tuned on it.
- Each configuration was trained once; we did not measure variance across random seeds, use cross-validation, or ensemble models.
- The mask head predicts at 28×28 per box, which limits the detail of thin parts such as barbs.
- Chirality labels are not used, and no step specifically targets fragmentation or over-merging.
- The human-agreement estimate uses pairs of training annotations and may not match how the test set is scored.
- Each configuration has a single public score, the first two models scored the same 0.34 despite different local scores, and the leaderboard may cover only part of the test set.

7 Conclusion

A pretrained Mask R-CNN with flip augmentation, a 1280 px input, PQ-tuned thresholds and disjoint-mask post-processing yields valid, fast predictions whose agreement with the labels matches the agreement between human annotators. Further gains are more likely to come from better recall on small filaments than from sharper outlines. Code, weights and the end-to-end notebook are available at `https://github.com/aayu-sha/MAGFiLO-Challenge-2026`.

References

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